# Combination circuits

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Reduce a circuit by identifying genuine series and parallel groups.

**SUCCESS CRITERIA - I can...**

1. Identify nodes before grouping resistors.
2. Reduce one group at a time.
3. Check current and voltage conservation in the original circuit.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

A 3 Ω resistor carries 5 A. Find the potential difference across it.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $R_{series} = R_1 + R_2$; $1/R_{parallel} = 1/R_1 + 1/R_2$; $V = IR$. Reduce one known group at a time. Check nodes before calling components series or parallel.

**Worked reasoning:** A 2 ohm resistor is in series with two 4 ohm resistors in parallel. The parallel pair equals 2 ohms; total 4 ohms. A 12 V supply gives 3 A total.

**Your turn:** Draw the network and label the two series sections before splitting the branch current.

**Marzano check - circle:** ① ② ③

## 03 Look and explain

A 2 Ω resistor is in series with two parallel 6 Ω resistors. Draw the arrangement and circle the parallel group.




Circuit symbols: long/short lines = battery; rectangles = resistors.

**Marzano check - circle:** ① ② ③

## 04 Build the solution

Use the supplied equivalent value: two parallel 6 Ω resistors act like 3 Ω. Add the series 2 Ω. With a 10 V battery, find total current.

**Marzano check - circle:** ① ② ③

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Continue down each column.

### 05 Use the idea

A student adds all three resistor values because they are drawn side by side. What should determine the grouping instead?

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A radar echo’s complete outward-and-return path is 6000 m. Find the target range and explain the division.

**Marzano check - circle:** 1 2 3

### 07 Exit check

If 2 A reaches two equal parallel branches, find each branch current.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

Return to the worked example in problem 2. Replace the parallel pair by two 8 ohm resistors. Find the new equivalent resistance and total current on 12 V.

**Marzano check - circle:** 1 2 3